ENGRO FERTILIZERS — DAHARKI

AMMONIA / UREA PLANT — LIVE-PLANT & CONSTRUCTION INTERFACE RISK ASSESSMENT

CONTROLLED DOCUMENT | ACTIVITY-BASED HAZARD AND RISK REGISTER | PUBLIC-SOURCE PRE-ASSESSMENT

DOCUMENT STATUS: NOT APPROVED FOR EXECUTION — SITE DATA, WORK PACK AND OWNER REQUIREMENTS MUST BE VERIFIED

Public-source pre-assessment — site validation and authorized approval required. This document follows a generic professional reference format for layout guidance; it is NOT an official Engro document, is NOT Engro-approved, and is not evidence of a site inspection.

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1. DOCUMENT CONTROL & PROJECT INFORMATION

Complete every applicable field with verified information. Do not invent missing site information.

Project / Facility Engro Fertilizers ammonia/urea complex, Daharki, Sindh, Pakistan — Base Plant and EnVen plant [CONFIRM exact unit/plot with approved plot plan]	**Client / Asset Owner** Engro Fertilizers Limited (operator) [S-01][S-02] [S-05]	**Principal Contractor** [SITE INPUT REQUIRED — not publicly confirmed]	**Subcontractor(s)** [SITE INPUT REQUIRED — not publicly disclosed]
Assessment Title / Activity Live-plant / construction, maintenance and turnaround interface risk assessment — activities only where confirmed in the approved work pack	**Work Location / Area** Daharki complex [CONFIRM unit/plot]	**RA Number / Document ID** ENG-DAH-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL]	**Revision / Status** Rev. 0 — Public-source pre-assessment — NOT APPROVED FOR EXECUTION
Assessment Date 28 October 2026 (public-source research snapshot)	**Review / Expiry Date** [PER OWNER PROCEDURE — set at approval; triggers in Section 15]	**Related PTW / JSA / Method Statement** [SITE INPUT REQUIRED — from site PTW register and approved work pack]	**Applicable Procedure / Standard** Owner HSE/PSM procedures [NOT REVIEWED — not public]; ISO 45001:2018+Amd 1:2024 framework [S-12]; Sindh Factories Act 2015 [S-06]
Prepared By / Role [REAL NAME / TITLE / SIGNATURE — authorized competent person]	**Reviewed By / Role** [REAL NAMES / TITLES — Operations, Construction, Process/Mechanical/Electrical, HSE]	**Approver / Role** [AUTHORIZED APPROVER ONLY — this Rev. 0 is NOT approved]	**Workforce / Shifts** Current site headcount NOT PUBLICLY DISCLOSED — obtain gate/mobilisation records (Section 5)

1.1 Contents

1. Document control & project information | 2. Executive summary | 3. Project profile and source-backed facts | 4. Site footprint, asset and workfront register | 5. Workforce and exposure register | 6. Assessment methodology and applicable requirements | 7. Detailed activity-based risk assessment register (26 entries) | 8. Critical controls / verification checklist | 9. Task sequencing / JSA / permit matrix | 10. Responsible-role matrix | 11. Action tracker | 12. Emergency response and rescue interface | 13. Environmental and occupational-health screening | 14. Document limitations, assumptions and site-data request list | 15. Approval, validation and review | 16. Quality check | 17. References and traceability

VALIDATION DISCLAIMER: Prepared from public sources only (Section 17, accessed 28 October 2026). No site visit, interview, permit/drawing review or headcount verification has taken place. Every unverified site-specific value is marked [CONFIRM] / [SITE INPUT REQUIRED].

2. Executive summary

Engro Fertilizers operates an ammonia/urea complex at Daharki, Sindh, comprising the Base Plant and the EnVen plant — described by the operator at its 2011 launch as the world's largest single-train ammonia-urea plant, at 1.3 million tons per annum [S-01]. The complex was operating in 2025: EnVen resumed on 7 June 2025 after a brief unscheduled-maintenance shutdown [S-02]. Construction, maintenance and turnaround work therefore interfaces with a live ammonia/urea process — and ammonia toxicity is the defining major-accident hazard of this process type.

Top hazards (Section 7): ammonia loss of containment and toxic exposure (RA-01/02); reformer/furnace failure (RA-03); high-pressure synthesis-loop release (RA-04); line opening (RA-10); confined-space entry (RA-11); SIMOPS breakdown at the live-plant interface (RA-18); and emergency-response failure in an ammonia release (RA-24). Catalyst handling (RA-07), process-gas composition (RA-08) and nitrogen use (RA-26) are VERIFICATION-CRITICAL: their controls cannot be finalised until SDS/composition/procedure data is received. Heat stress in Sindh summer conditions (RA-21) is a critical, legally grounded hazard — Pakistan jurisdiction applies (Sindh factory/OSH law [S-06]); there is no Saudi-style statutory midday ban to copy.

Hold points. Work must not start until, as a minimum: ammonia inventory, P&IDs and detector coverage are verified; gas/SDS data is received and the RA-07/08 decisions recorded; the owner risk matrix replaces the PROPOSED matrix in Section 6; the approved emergency plan (alarm, upwind muster, headcount) is briefed to every worker; and actual shift headcount capture is running. Residual risk is NOT declared acceptable by this pre-assessment.

3. Project profile and source-backed facts

Fact	Value / statement	Status & source	What it does — and does not — prove
Operator / project	Engro Fertilizers Limited — ammonia/urea complex at Daharki, Sindh, Pakistan. Subsidiary of Engro Corporation [S-02]; GEM lists Engro Fertilisers Limited as 100% owner/operator [S-05].	Verified public fact [S-01][S-02][S-05]	Establishes the real operator and site. Does NOT establish the exact assessment plot boundary — plot plan required.
Products / feedstock	Primary product ammonia; downstream urea; feedstock natural gas [S-05].	Verified public fact [S-05]	Confirms the process family this RA addresses. Does NOT give inventories, pressures or compositions.
EnVen plant	Commissioned/established 2011; USD 1.1bn; single-train ammonia-urea; designed capacity 1.3 million tons urea/yr; described at launch as the world's largest single-train ammonia-urea plant and Pakistan's most energy-efficient fertilizer plant [S-01][S-02].	Verified public fact [S-01][S-02]	Design capacity as publicly reported. NOT today's measured production rate and NOT process design data.
Base Plant	Daharki Plant-1 capable of up to 950,000 tons urea/yr after maintenance/debottlenecking, per a PSX notice reported Dec 2022 (previously 780,000 t, before that 500,000 t) [S-04].	Verified public fact [S-04] — see note	One secondary summary states 975,000 t / total 2.275 Mt [S-04]; this RA uses the PSX-notice figure (950,000 t) and flags the discrepancy for operator confirmation.
Excluded site	Zarkhez NPK plant is at Port Qasim, Karachi — a separate location [S-04].	Verified public fact [S-04]	Excluded from this Daharki assessment, exactly as Riyas was excluded from the Jafurah assessment.
Operating status	EnVen resumed operations 7 June 2025 after unscheduled maintenance (shut 5 June 2025) [S-02].	Verified public fact [S-02]	Proves operation in mid-2025. Current October 2026 operating status must be confirmed with the operator — not assumed.
2024 turnaround (historical)	55-day EnVen turnaround completed 18 June 2024 (~USD 50m): >5,000 activities incl. waste-heat-boiler replacement, furnace convection overhaul, ammonia storage tank inspection, overhaul of six turbo trains/gas turbines; >6,000 people at peak [S-03].	Historically reported [S-03]	Proves turnaround scope types and scale in 2024. Does NOT prove today's workfronts, equipment condition or headcount.
Ammonia toxicity (process type)	Ammonia is corrosive/toxic: NIOSH REL 25 ppm TWA, IDLH 300 ppm; OSHA PEL 50 ppm TWA [S-07][S-08].	Technical guidance — US values, NOT Pakistan law	Characterises the hazard credibly. Site alarm/action limits must come from the approved site standard — never from this table.
Heat law (jurisdiction)	Sindh Factories Act 2015: ventilation/reasonable temperature, cooled drinking water in hot months for larger factories, rest provisions, hours limits [S-06].	Pakistan / Sindh LAW — topic level	Sets the jurisdictional frame for RA-21. Exact provisions to be confirmed by the site legal/HSE function. No foreign midday ban applies.

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THINNESS NOTE: Public information on Daharki is materially thinner than for a listed-energy megaproject — no public plot plan, P&ID set, inventory, detector layout, contractor register or current headcount exists. This pack is therefore deliberately verification-heavy; that is the defensible position, not filler.

4. Site footprint, asset and workfront register

Public sources confirm the units below at complex level only. Every operational field is [SITE INPUT REQUIRED]. Zarkhez (Port Qasim) is listed only to record its exclusion.

Facility / asset / workfront	Location	Package / contractor	Current activity & SIMOPS interface	Operational status	Source / drawing reference	Data owner	Verification status
Base Plant (Plant-1) — ammonia/urea	Daharki [CONFIRM plot/unit IDs]	[CONFIRM]	[CONFIRM — live operation assumed for interface purposes only]	Operating historically; current status [CONFIRM]	Capacity per PSX notice [S-04]; plot plan [SITE INPUT REQUIRED]	Operations	Not verified — site input required
EnVen plant — ammonia/urea	Daharki [CONFIRM]	[CONFIRM]	Construction/maintenance interface where work pack confirms	Resumed 7 Jun 2025 [S-02]; current [CONFIRM]	Operator release [S-01]; PSX report [S-02]	Operations	Status 2025 verified; current not verified
Ammonia storage & transfer area	Daharki [CONFIRM tank IDs]	[CONFIRM]	Storage/loading interface; tank inspection evidenced in 2024 scope [S-03]	In service [CONFIRM]	Turnaround report [S-03] (historical scope)	Mechanical	Not verified
Reformer, synthesis loop, utilities, steam/WHB	Daharki [CONFIRM]	[CONFIRM]	Process areas — see RA-03/04/19/20	In service [CONFIRM]	Process type + 2024 scope [S-03]	Process	Not verified
Urea prilling/granulation, bagging, warehouse/despatch	Daharki [CONFIRM configuration]	[CONFIRM]	Product handling interface	In service [CONFIRM]	Process type; site data required	Urea Plant	Not verified
Fabrication / laydown / turnaround areas	Daharki [CONFIRM]	[CONFIRM]	Prefabrication, storage, turnaround staging; traffic interface	[CONFIRM]	Work pack [SITE INPUT REQUIRED]	Construction Manager	Not verified
Zarkhez NPK plant — PORT QASIM, KARACHI	Karachi — SEPARATE SITE	n/a	NONE — excluded	Separate facility [S-04]	[S-04]	n/a	EXCLUDED — separate assessment required

5. Workforce and exposure register

5.1 Public figures — with date, scope and correct use

Figure	Value	Date / scope	Correct use in this assessment
Engro Fertilizers company employees	3,031 (2024, as reported in company financial summaries)	Company-wide, 2024 — all sites/functions	Corporate context ONLY. Not a Daharki site headcount; excludes contractor/turnaround personnel.
EnVen turnaround peak workforce (2024)	More than 6,000 people at peak, round-the-clock	55-day turnaround, Apr–Jun 2024 [S-03] — HISTORICAL, event-specific	Proves turnaround-scale exposure historically. NOT current staffing and NOT a routine-operations headcount.
Total persons inside the Daharki boundary on the issue date	NOT PUBLICLY DISCLOSED	Current — not found in any public source reviewed	Obtain from gate/access-control and contractor mobilisation/daily manpower records. Do not estimate.
Active workfronts / package count on the issue date	NOT CONFIRMED IN PUBLIC SOURCES	Current	Obtain from the approved plot plan, package list and permit register.

WARNING: a company-wide employee total and a one-off 2024 turnaround peak measure different things. Neither is the number of people inside the Daharki fence today, and neither may be used for muster, welfare, transport or emergency planning.

5.2 Actual manpower and exposure register — TO BE COMPLETED FROM SITE RECORDS

Source of count required: gate access-control/badge report, muster roster, contractor mobilisation register or daily manpower report. Until a dated record is attached, every field remains 'Not publicly disclosed — obtain actual site record.'

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Field	Value (from site record only)	Record reference
Headcount date/time; shift pattern (day/night)	Not publicly disclosed — obtain actual site record	
Direct operator personnel — day / night	Not publicly disclosed — obtain actual site record	
EPC/main contractor personnel — day / night	Not publicly disclosed — obtain actual site record	
Each subcontractor separately — day / night	Not publicly disclosed — obtain actual site record	
Vendor/OEM/commissioning personnel; visitors	Not publicly disclosed — obtain actual site record	
TOTAL persons inside the defined boundary	Not publicly disclosed — obtain actual site record	
Peak concurrent workforce (reporting period)	Not publicly disclosed — obtain actual site record	
Work-at-height personnel; confined-space entrants/attendants	Not publicly disclosed — obtain actual site record	
Lifting crew; hot-work crew/fire watches; E&I crew	Not publicly disclosed — obtain actual site record	
Emergency-response personnel on shift	Not publicly disclosed — obtain actual site record	
Prepared by / Verified by (name/role); revision & period	Not publicly disclosed — obtain actual site record	

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6. Assessment methodology and applicable requirements

Method. Work is decomposed into activity steps (Section 7). For each: hazard source and specific hazardous event, persons exposed and how, credible consequences including fatality/major-accident outcomes, existing controls with evidence status, additional controls in hierarchy order (elimination, substitution, engineering, administrative, PPE), responsible role, and residual rating only for controls that are implemented and verified. A rating is never reduced by writing 'follow procedure' or adding PPE alone.

Three risk values. Initial/inherent (no controls beyond the task itself); current (only verified existing controls credited — in this draft, almost none are credited, because no site evidence was reviewed); residual (after additional controls are implemented and competently checked). All ratings here are PROVISIONAL professional judgements for prioritisation, to be re-rated in the site validation workshop with the owner matrix.

Major-accident limitation. A 5x5 matrix screens and prioritises; it does not assess major-accident scenarios alone. A low likelihood score must never dismiss a catastrophic ammonia scenario (RA-01/02/04). Such scenarios require the owner's specialist studies (HAZID/HAZOP/LOPA/QRA, bow-tie) — an interface listed as a site input in Section 14. This is an activity-based HSE risk assessment, not a process hazard analysis, method statement or task JSA.

6.1 Proposed 5 x 5 risk matrix — PROPOSED WORKING MATRIX ONLY

Use the owner-approved matrix whenever one exists. The matrix below is PROPOSED only and must be replaced or formally accepted before use. Score = Likelihood x Severity.

PROPOSED 5 x 5 RISK MATRIX

S x L	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

RATING DEFINITIONS (TO BE APPROVED)

LIKELIHOOD

Level	Term	Definition / guide
1	Rare	Exceptional; highly unlikely in the defined conditions.
2	Unlikely	Not expected, but credible.
3	Possible	Could occur under some circumstances.
4	Likely	Expected in many circumstances.
5	Almost certain	Expected frequently/repeatedly.

SEVERITY

Level	Term	Definition / guide
1	Insignificant	No injury or negligible harm/damage.
2	Minor	First-aid case or minor reversible damage.
3	Moderate	Medical-treatment/lost-time potential; significant reversible harm.
4	Major	Single-fatality potential, life-changing injury or major damage.
5	Catastrophic	Multiple fatalities or catastrophic asset/environmental consequences.

PROPOSED BANDS (subject to owner approval): 1–4 Low — routine controls. 5–9 Moderate — specific controls and supervision. 10–16 High — additional controls, documented review and authorized acceptance before work. 17–25 Extreme — do not proceed until risk is reduced and acceptance is given at the level the owner's system requires. Not a universal or owner matrix.

6.2 Applicable requirements and verification register

Requirement / source	Jurisdiction & type	Relevant topic (summarised)	Applicability	Version/date checked	Owner document still needed	Verification status
Sindh Factories Act 2015; Sindh OSH Act/Rules [S-06]	Pakistan / Sindh — LAW	Ventilation and reasonable temperature; cooled drinking water in hot months (larger factories); rest provisions; hours limits	All factory work	Act XIII of 2016 as amended; checked 28 October 2026	Site legal register with exact current provisions	Topic verified; exact provisions to confirm
Pakistan environmental / factory licensing framework	Pakistan — LAW	Emissions, effluent and waste duties at topic level only	Plant operations	Not verified in this check	Project legal register; SEPA consents	SITE INPUT REQUIRED —

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Requirement / source	Jurisdiction & type	Relevant topic (summarised)	Applicability	Version/date checked	Owner document still needed	Verification status
(general)					[CONFIRM]	no clauses cited from memory
ISO 45001:2018 + Amd 1:2024 [S-12]	International — STANDARD	OH&S framework: hazard identification, participation, operational control, emergency preparedness	Assessment process	Confirmed 2024; checked 28 October 2026	Project HSE plan mapping	Verified as framework only
Owner (Engro) HSE / PSM / PTW requirements	Client-specific — CONTRACTUAL	PTW, isolation, ammonia response, lifting, confined space, competency	All work — governing for execution	Not public; NOT reviewed	Authorized copies via document control	NOT REVIEWED — critical site input
NIOSH / OSHA ammonia and PSM guidance [S-07][S-08][S-09]	United States — GUIDANCE ONLY	Ammonia toxicity values; PSM concepts for interface work	Technical reference RA-01/04/08/18	Checked 28 October 2026	Owner PSM standard (governing)	Guidance only — NOT Pakistan law
OSHA confined-space / welding guidance [S-10][S-11]	United States — GUIDANCE ONLY	Control concepts for entry and hot work	Technical reference RA-11/12	As cited 28 October 2026	Owner procedures (governing)	Guidance only — NOT Pakistan law

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7. Detailed activity-based risk assessment register

26 specific entries. Ratings use the PROVISIONAL matrix in Section 6.1 and are professional judgements for prioritisation only. Existing controls carry evidence status; a control without evidence is NOT credited until verified on site.

RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
RA-01	Work on/near ammonia production, transfer and process piping	Ammonia plant areas, Daharki [CONFIRM unit/plot]	Loss of containment of anhydrous ammonia from piping, vessels, pumps/compressors or flanges, forming a toxic gas cloud. Ammonia toxicity is a documented hazard of the process type: corrosive to eyes, skin and respiratory tract; US NIOSH IDLH 300 ppm, REL 25 ppm TWA; OSHA PEL 50 ppm TWA — TECHNICAL GUIDANCE values only, not Pakistan law [S-07][S-08]. Site inventory, pressures and detector coverage are NOT public.	Acute toxic exposure, chemical burns, blindness, lung injury; single or multiple fatalities credible; off-site/community impact possible in a major release. Persons at risk: Process and maintenance crews at the release point and downwind personnel; exposure via inhalation and skin/eye contact.	TO BE VERIFIED: fixed/portable ammonia detection, ESD/isolation, emergency showers and owner PSM arrangements not evidenced — site input required.	3 x 5 = 15 — HIGH	VERIFY FIRST: obtain P&IDs, ammonia inventory, operating pressures and detector layout. ENGINEERING: positive isolation, verified gas-detection coverage, remote isolation where fitted. ADMIN: ammonia-specific permit controls, upwind work positioning, personal ammonia monitors with alarm limits set ONLY to the approved site standard, rehearsed release response. PPE: ammonia-rated respiratory/eye/skin protection per site standard (last line only).	Process Safety / HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner PTW; isolation certificate; ammonia work authorisation [SITE INPUT REQUIRED] Review trigger / reference: Review on any boundary/inventory change. Refs: NIOSH/OSHA ammonia guidance [S-07] [S-08] (guidance only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no intrusive ammonia work until inventory/P&ID/detector data received and release-response arrangements verified. Residual HIGH — owner acceptance only.
RA-02	Ammonia storage tank inspection, maintenance and loading/transfer operations	Ammonia storage area [CONFIRM tank IDs/capacity — not public]	Tank or transfer-system failure, overfill, hose/loading-arm failure or vapour release during inspection/transfer. Public reporting confirms an ammonia storage tank inspection formed part of the 2024 EnVen turnaround scope [S-03]; tank capacity, condition and safeguards are NOT public.	Large toxic release, multiple fatalities credible, major environmental/community impact. Persons at risk: Tank/loading crews, nearby plant personnel, potentially off-site receptors downwind.	TO BE VERIFIED: tank inspection status, relief/level safeguards, bunding/drainage and transfer procedures not evidenced.	2 x 5 = 10 — HIGH	ENGINEERING: verified high-level/relief protection, transfer-system inspection before use, earthing/bonding. ADMIN: written transfer procedure with independent level verification, exclusion zone sized by site procedure, continuous attendance, weather/wind check, emergency isolation point briefed. PPE: per RA-01.	Mechanical Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Tank entry/transfer permits; inspection records [SITE INPUT REQUIRED] Review trigger / reference: Review per inspection cycle. Ref: Business Recorder turnaround report [S-03] (scope only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no transfer or tank work without verified safeguards and a briefed emergency-isolation route.
RA-03	Primary reformer / furnace operation and maintenance (incl. convection section)	Reformer area [CONFIRM]	Furnace tube failure, fuel-gas leak, burner/flame failure, refractory failure or hot-surface contact. Public reporting confirms overhaul of the furnace convection section and waste-heat-boiler replacement in the 2024 turnaround	Fire/explosion, severe burns, fatality, major asset damage and escalation to adjacent process equipment. Persons at risk: Operations, maintenance and turnaround crews; exposure via flame, radiant heat and released fuel	TO BE VERIFIED: burner management, tube inspection history, gas detection and furnace safeguards not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: verified burner-management/flame-failure protection, tube integrity inspection per owner programme. ADMIN: furnace light-up/purge procedure by authorised operators only, hot-surface guarding/signage, gas testing before maintenance entry into convection sections. PPE: heat/flame protection per task.	Process Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Furnace work/light-up procedure; PTW [SITE INPUT REQUIRED] Review trigger / reference: Review on any tube/burner finding. Ref: [S-03] (historical scope only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no furnace light-up or internal work without authorised procedure, purge verification and operations sign-off.

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
			[S-03]; current furnace condition is NOT public.	gas/process gas.							
RA-04	Work on the high-pressure ammonia synthesis loop and converters	Synthesis loop [CONFIRM pressures from P&IDs — no values assumed]	High-pressure gas release (hydrogen/ammonia/ syngas mixture per process design — composition to be confirmed), jet fire, embrittlement/material failure or flange/joint failure under pressure.	Jet fire/explosion, toxic exposure, fatality, major plant damage. Persons at risk: Crews at the work point and personnel in jet-fire/release zones; exposure via thermal radiation, blast and toxic gas.	TO BE VERIFIED: design/operating pressures, materials/inspection records and loop isolation provisions not public.	3 x 5 = 15 — HIGH	VERIFY FIRST: P&IDs, pressures, materials and inspection status. ENGINEERING: positive isolation and proven depressurisation before break-in. ADMIN: joint-integrity controls (RA-10/RA-12 principles), staged repressurisation with leak survey, exclusion zone during first pressurisation. PPE: per RA-01 plus FR clothing.	Mechanical Engineer / Process Safety Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; leak-test record Review trigger / reference: Review on any loop modification. Ref: OSHA PSM guidance [S-09] (guidance only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no break-in until depressurisation is proven at the work point and recorded; no repressurisation without leak test and operations acceptance.
RA-05	Natural-gas feed, fuel and utility gas systems — work and tie-ins	Gas feed/fuel systems [CONFIRM]	Flammable gas release from feed/fuel piping, valves or tie-in work; ignition forming jet fire/flash fire. Natural gas is the confirmed feedstock for the Daharki ammonia plant [S-05]; site pressures/routes are NOT public.	Fire/explosion, fatalities, escalation to reformer/process areas. Persons at risk: Work crews and operators; exposure via thermal radiation and blast.	TO BE VERIFIED: gas routing, isolation points, detection and tie-in procedures not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: positive isolation, purge and gas-free verification before hot/tie-in work. ADMIN: tie-in procedure with independent lineup verification, leak test before introduction of gas, ignition-source control in the work zone. PPE: FR clothing per site standard.	Area Authority / Mechanical Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Tie-in permit; isolation certificate; leak-test record Review trigger / reference: Review on any lineup change. Ref: GEM plant entry (feedstock) [S-05].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no gas introduction without leak test, purge records and operations acceptance.
RA-06	Urea prilling/granulation, melt handling and product conveying	Urea plant [CONFIRM prilling vs granulation configuration at each unit]	Molten-urea contact burns; urea/ammonia dust exposure; slips from product spillage; entanglement in conveyors/elevators; dust build-up and housekeeping failures.	Thermal burns, respiratory/eye irritation, entanglement injuries, lost-time injuries. Persons at risk: Urea plant operators and maintenance crews; exposure via contact, inhalation and moving machinery.	TO BE VERIFIED: unit configuration, dust controls, guarding and exposure monitoring not evidenced.	4 x 3 = 12 — HIGH	ENGINEERING: machine guarding and conveyor interlocks verified, local exhaust/dust control, melt-line insulation and splash protection. ADMIN: lockout for conveyor work (RA-09), spillage clean-up standard, dust exposure assessment and monitoring where indicated. PPE: heat gloves/face protection for melt work; dust respirator selected from exposure assessment.	Urea Plant Supervisor / HSE Officer Priority: High Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	PTW; LOTO records; hygiene survey [SITE INPUT REQUIRED] Review trigger / reference: Review on monitoring results. Refs: process-type hazards; site data required.	Evidence status: To be verified — no site evidence reviewed STOP-WORK: unguarded conveyor or failed dust control stops the affected task until corrected.
RA-07	Catalyst handling: reformer / shift / synthesis catalyst change-out — VERIFICATION-CRITICAL	Converter/ reformer vessels [CONFIRM catalyst types from owner data]	Exposure to catalyst dust/materials whose composition is NOT public (some process catalysts are toxic, pyrophoric when spent, or hazardous on air contact — type-specific SDS required before any assumption); oxygen-deficient vessel atmosphere during change-out.	Occupational illness, chemical burns, asphyxiation; severity depends entirely on the actual catalyst — not assumed here. Persons at risk: Change-out crews and vessel entrants; exposure via dust inhalation/skin contact and vessel atmosphere.	NOT CONFIRMED: catalyst types, SDS and change-out procedure were not available to this pre-assessment.	3 x 4 = 12 — HIGH	VERIFY FIRST: catalyst list + current SDS + owner change-out procedure before mobilisation. ENGINEERING: ventilation, dust suppression/extraction, inert-atmosphere controls where the procedure requires. ADMIN: catalyst-specific risk assessment per SDS, trained change-out crew, confined-space controls (RA-11) for vessel entry, spent-catalyst handling/disposal per SDS and environmental plan. PPE: strictly per SDS.	Process Engineer / HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 4 = 4 — LOW	Catalyst SDS file; change-out procedure; entry permit Review trigger / reference: Review on any catalyst change. Ref: SDS [SITE INPUT REQUIRED before use].	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no catalyst handling before SDS-based assessment is complete and briefed to the crew.
RA-	Work where carbon	Reformer/	Toxic process-gas	Acute poisoning,	NOT CONFIRMED: no	3 x 5 =	VERIFY FIRST: obtain stream	Process Safety /	2 x 5 =	Gas test	Evidence status:

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
08	monoxide / process gas exposure may occur — VERIFICATION-CRITICAL	shift/ synthesis areas [CONFIRM from process data]	exposure (e.g. CO-bearing syngas streams are inherent to ammonia production by reforming as a process type); exact stream compositions at Daharki are NOT public.	fatality; failed rescue multiplying casualties. Persons at risk: Crews opening or working near process systems; exposure via inhalation at the work point.	gas composition, P&ID or SDS data available to this pre-assessment.	15 — HIGH	composition data and owner gas-testing criteria. ENGINEERING: fixed/portable detection coverage as process data requires. ADMIN: authorised gas testing before/during intrusive work, personal monitors with limits ONLY per approved site standard, buddy system and rescue plan. PPE: respiratory protection per site procedure.	HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED]	10 — HIGH	certificate; owner gas-testing procedure [SITE INPUT REQUIRED] Review trigger / reference: Review immediately when composition data is issued. Ref: OSHA PSM hazard recognition [S-09] (guidance only).	NOT CONFIRMED — verification-critical HOLD POINT: intrusive work prohibited until composition data received and the gas-testing decision recorded by a competent person.
RA-09	Isolation of mechanical/electrical/instrument energy and lockout-tagout (LOTO)	All systems in package [CONFIRM isolation points from P&IDs]	Failure to identify/isolate all energy sources (pressure, electrical, stored, gravity, chemical) causing unexpected start-up or release during intrusive work.	Fatality or serious injury from pressure release, moving equipment or electric shock. Persons at risk: Fitters, electricians and instrument technicians at the work point.	TO BE VERIFIED: owner LOTO/isolation procedure and isolation hardware not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: positive isolation (physical disconnection/blinding) for line breaking per owner procedure; lockable points. ADMIN: isolation certificate listing every energy source, personal locks/tags, zero-energy verification (try-start, test-for-dead, bleed check) recorded before work, group lockout for multi-craft work.	Authorised Isolator / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Isolation certificate; LOTO register; PTW Review trigger / reference: Review on any boundary change. Ref: owner isolation procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: work must not start until zero-energy state is proven and recorded; any doubt stops the job.
RA-10	Line opening / flange breaking on process piping	Tie-in and modification points [CONFIRM from line list]	Residual pressure, trapped gas/liquid (incl. ammonia) or wrong-line identification releasing process fluid at the break point; line-of-fire ejection.	Toxic/flammable release, fire, fatality, environmental release. Persons at risk: Pipefitters at the flange face and personnel in the line of fire.	TO BE VERIFIED: owner line-opening procedure, drain/vent provisions and lineup drawings not reviewed.	4 x 5 = 20 — EXTREME	ELIMINATION: break only at designated points in the approved procedure. ENGINEERING: drain, vent, purge and prove depressurised at the break point; first-break technique from a protected position. ADMIN: two-person line identification against P&ID/line list, gas test at break point, operations-signed permit. PPE: face/chemical protection per fluid SDS and site standard.	Mechanical Supervisor / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; gas test certificate Review trigger / reference: Review for every line break. Ref: owner procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: first break prohibited until drain/vent proven, gas test satisfactory and permit signed. Stop on any unexpected pressure/product.
RA-11	Confined-space entry: vessels, columns, tanks, pits and sumps	Equipment requiring internal work [CONFIRM entry list]	Toxic/flammable/ oxygen-deficient atmosphere (incl. ammonia or inert-gas residues), engulfment or heat stress inside the space; failed rescue multiplying casualties.	Single or multiple fatalities (entrant and would-be rescuers). Persons at risk: Entrants, attendant and rescue team; exposure via atmosphere inside the space.	TO BE VERIFIED: owner confined-space procedure, classification and rescue capability not evidenced.	3 x 5 = 15 — HIGH	ELIMINATION: remote inspection/cleaning where practicable. ENGINEERING: isolation/blinding of connected lines, cleaning/purging proven by test, continuous ventilation and monitoring. ADMIN: entry permit, named attendant, communications check, entry log, rehearsed rescue plan with equipment at the entry point before entry. PPE: respiratory protection per test results/site procedure.	Confined-Space Entry Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Confined-space entry permit; gas test certificate; rescue plan Review trigger / reference: Review for every entry. Ref: OSHA confined spaces guidance [S-10] (guidance only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no entry without rescue capability physically in place; entry stops on alarm, ventilation failure or attendant leaving post.
RA-12	Hot work: welding, cutting and grinding in/near process areas	Fabrication areas and in-situ locations [CONFIRM]	Sparks/slag igniting flammable atmosphere, combustibles or process residues;	Fire/explosion, fatalities, burns, eye injury and fume illness. Persons at risk: Hot-work crew, fire watch	TO BE VERIFIED: owner hot-work procedure, gas-testing criteria and fire-watch	3 x 5 = 15 — HIGH	ELIMINATION: prefabricate in a designated safe area where possible. ENGINEERING: spark containment, removal/isolation of combustibles within the	Hot Work Supervisor / Fire Watch Priority: Critical Target date:	1 x 5 = 5 — MODERATE	Hot-work permit; gas test certificate; fire-watch record Review trigger /	Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
			fire spreading beyond the work point; welding fume and arc radiation.	and nearby workers/plant.	practice not reviewed.		radius set by the site procedure, fume extraction in enclosed areas. ADMIN: hot-work permit, gas testing where required, trained fire watch during work and for the post-work period in the site procedure, verified extinguishers at the work point. PPE: welding PPE per site standard.	[SITE INPUT REQUIRED]		reference: Review for every permit. Ref: OSHA welding guidance [S-11] (guidance only).	work stops on gas-test alarm, loss of fire watch or any plant upset alarm in the area.
RA-13	Crane lifting of equipment, spools and turnaround components	Laydown and installation points [CONFIRM]	Rigging failure, wrong load weight/centre of gravity or crane malfunction causing a dropped/swinging load; lifts over live ammonia/process equipment escalate consequences.	Fatality, serious injury, damage to live plant with secondary ammonia/hydrocarbon release. Persons at risk: Rigging crew and personnel in the drop zone.	TO BE VERIFIED: lifting procedure, crane/rigging certification and appointed-person arrangements not reviewed. The 2024 turnaround (>5,000 activities [S-03]) shows lift intensity in turnaround conditions — historical context only.	3 x 5 = 15 — HIGH	ENGINEERING: certified crane/rigging matched to a calculated lift plan (weights from vendor/drawing data — never estimated); tag lines. ADMIN: lift categorisation per owner standard, approved plan and appointed person for complex/critical lifts, barricaded drop zone, trial lift, no personnel under suspended loads. Ground-bearing verification for set-up positions.	Lifting Supervisor / Appointed Person Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved lift plan; crane/rigging certificates; lifting permit Review trigger / reference: Review for every complex/critical lift. Ref: owner lifting procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: critical lifts over/near live plant require operations approval in addition to the lift plan.
RA-14	Work at height, scaffolding and dropped objects	Elevated areas incl. reformer/urea structures [CONFIRM]	Fall from height through open edges/penetrations; scaffold collapse; tools/materials dropped onto personnel below.	Fatality or life-changing injury to the faller or the person struck. Persons at risk: Fitters, welders, inspectors at height; all personnel below.	TO BE VERIFIED: edge protection, anchor certification, scaffold system and rescue arrangements not evidenced.	4 x 5 = 20 — EXTREME	ELIMINATION: pre-assemble at grade where practicable. ENGINEERING: collective protection first — guardrails, covered penetrations, certified anchors; tool tethering, toe boards, barricaded drop zones. ADMIN: scaffolds erected/inspected/tagged by competent scaffolders; 100% tie-off per site standard; rescue-from-height plan and equipment at the workfront before work starts. PPE: full-body harness with double lanyard per site standard.	Construction Supervisor / Scaffold Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Work-at-height permit (if site system requires); scaffold register; rescue plan Review trigger / reference: Review for every new elevated workfront. Ref: owner procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: work at unprotected edges prohibited; red-tagged scaffolds physically closed.
RA-15	Excavation and trenching; buried live services	Civil areas [CONFIRM]	Trench collapse; strike on buried live services (gas, ammonia, electrical, water) not shown or wrongly marked on drawings.	Burial/asphyxiation fatality; toxic/flammable release or electrocution from a service strike. Persons at risk: Workers in/beside the excavation; plant operators.	TO BE VERIFIED: underground-service drawings, permit practice and soil data not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: protective system selected by a competent person for actual soil/depth — dimensions from the excavation design, not assumed; service detection scan and trial holes before mechanical digging. ADMIN: excavation permit with services clearance signed by operations/E&I, daily and after-rain inspections, mechanical digging stops at the agreed distance from marked services. PPE: task PPE.	Civil Supervisor / Competent Person (Excavations) Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Excavation permit; services-clearance drawings; scan records Review trigger / reference: Review daily and after rain. Ref: owner excavation procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no mechanical excavation without signed services clearance; any unmarked service found stops work in that excavation.
RA-16	Electrical work, temporary power and arc flash	Substations, MCCs, field installations [CONFIRM voltages from drawings —	Contact with live parts, backfeed from temporary supplies, damaged cables in dusty/wet conditions, arc flash during	Electrocution, arc-flash burns, fatality, fire. Persons at risk: Electricians, instrument technicians and all users of	TO BE VERIFIED: electrical safety rules, authorised-person register, drawings and temporary-power inspection regime not	3 x 5 = 15 — HIGH	ENGINEERING: isolation at all sources incl. temporary supplies; RCD/earth protection and protected cable runs. ADMIN: authorised electrical persons only, prove-dead recorded before contact,	Authorised Electrical Person / E&I Engineer Priority: Critical Target date: [SITE INPUT	1 x 5 = 5 — MODERATE	Electrical permit; isolation certificate; inspection register Review trigger / reference:	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no contact work until prove-dead

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
		none stated here]	switching/testing.	portable equipment.	reviewed.		electrical permit for HV work as defined by site rules, inspection/tagging of portable tools with defective items removed. PPE: arc-rated clothing and voltage-rated gloves per site electrical standard.	REQUIRED]		Review on any supply change. Ref: owner electrical rules [SITE INPUT REQUIRED].	recorded; any unidentified cable is treated as live and stops the job.
RA-17	Hydrostatic / pneumatic pressure testing of piping and equipment	Test boundaries [CONFIRM test pressures from approved test pack — none stated here]	Failure of a test blank, hose or weak component under pressure releasing high-energy jet/projectiles; pneumatic testing carries far higher stored energy than hydrostatic.	Fatality or serious injury; equipment damage; multiple fatalities credible in pneumatic failure. Persons at risk: Test crew and anyone inside/near the test boundary.	TO BE VERIFIED: test procedures, calibrated-gauge records and boundary practice not reviewed.	3 x 5 = 15 — HIGH	ELIMINATION/ SUBSTITUTION: hydrostatic testing wherever the specification allows; pneumatic only with engineered justification and senior approval. ENGINEERING: approved test pack (medium, pressure, hold points), restraints on temporary connections, calibrated gauges. ADMIN: barricaded exclusion zone, staged pressurisation, single command, controlled depressurisation. PPE: task PPE outside the line of fire.	Mechanical / Commissioning Engineer Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved test pack; pressure-test permit; calibration certificates Review trigger / reference: Review for every test. Ref: approved test procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: pressurisation only with the boundary confirmed clear and recorded; pneumatic testing must not proceed without its specific engineered review.
RA-18	SIMOPS: construction/turnaround work interfacing with the live ammonia/urea plant	Live-plant interface areas [CONFIRM]	Conflicting permits/activities (hot work near ammonia operations, simultaneous line opening and production) creating pathways no single permit holder can see; impaired detection/ESD left bypassed for convenience.	Major toxic/flammable release, multiple fatalities, conflicting emergency actions. Persons at risk: All work parties in shared areas; exposure via shared atmosphere, escape routes and permits.	TO BE VERIFIED: owner SIMOPS procedure, permit register and bypass/impairment register not reviewed. Turnaround scale is evidenced historically: 55-day 2024 EnVen turnaround, >5,000 activities, >6,000 people at peak [S-03] — NOT a current headcount.	3 x 5 = 15 — HIGH	ENGINEERING: single permit-coordination point with a live permit map for the area. ADMIN: SIMOPS matrix of prohibited combinations approved by operations, one accountable Area Authority per area/shift, bypasses only under formal MOC/impairment permit with compensating measures and restoration verification, permit conflict check at issue and every handover. PPE: per task RA.	Area Authority / Operations Representative Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner PTW system; SIMOPS plan; impairment register Review trigger / reference: Review at each shift change. Ref: owner PTW/SIMOPS procedure [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: hot work prohibited during ammonia venting/draining/line opening in the same area unless a specific SIMOPS assessment signed by operations authorises it.
RA-19	High-pressure steam, waste-heat boiler and condensate systems	Boiler/steam areas [CONFIRM]	Steam/condensate release, water hammer, boiler/WHB integrity failure or scalding during maintenance and inspection. WHB replacement and inspection formed part of the 2024 turnaround scope [S-03]; current condition NOT public.	Severe scalds, fatality, equipment damage. Persons at risk: Boiler/maintenance crews and nearby personnel.	TO BE VERIFIED: boiler inspection status, steam traps/relief protection and isolation provisions not evidenced.	3 x 4 = 12 — HIGH	ENGINEERING: verified relief protection and inspection per owner/statutory programme (regime to be confirmed — not assumed). ADMIN: proven isolation, cooling and drain-down before break-in, warm-up/pressurisation procedure by authorised operators, lagging/hot-surface protection. PPE: heat/steam protection per task.	Mechanical Engineer / Area Authority Priority: High Target date: [SITE INPUT REQUIRED]	1 x 4 = 4 — LOW	Boiler inspection records; PTW; isolation certificate Review trigger / reference: Review per inspection cycle. Ref: [S-03] (historical scope only).	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no break-in until isolation, cooling and zero pressure are proven and recorded.
RA-20	Rotating equipment: turbo trains, compressors, pumps and gas turbines	Compressor/ turbine areas [CONFIRM]	Unexpected start-up during maintenance, coupling/guard failure, or overspeed/mechanical failure ejecting fragments. Overhaul of six major turbo trains and gas turbines formed part	Fatality or serious injury from rotating parts, fragments or released process fluid. Persons at risk: Maintenance crews at the machine; nearby personnel in fragment/release	TO BE VERIFIED: machine guarding, overspeed protection, LOTO provisions and maintenance procedures not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: guards refitted before any rotation test; verified overspeed/protective systems per owner programme. ADMIN: LOTO with try-start verification (RA-09), permit cross-check before energisation/rotation, exclusion during first run after overhaul, vibration/condition monitoring per owner	Mechanical Supervisor / Commissioning Lead Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	LOTO records; overhaul/test records [SITE INPUT REQUIRED] Review trigger / reference: Review after every overhaul. Ref: [S-03]	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no energisation/rotation of any machine with open permits on that machine — boundary check

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
			of the 2024 turnaround [S-03]; current fleet condition NOT public.	zones.			programme. PPE: task PPE; no loose clothing near rotating equipment.			(historical scope only).	must confirm closure first.
RA-21	Heat stress during outdoor work in Sindh summer conditions	All outdoor workfronts, Daharki	Heat exhaustion progressing to heatstroke during high-temperature work; Daharki experiences extreme summer heat, and the 2024 turnaround was publicly reported as completed 'despite extreme weather conditions' [S-03]. Pakistan has NO Saudi-style statutory midday ban: applicable duties arise under the Sindh Factories Act 2015 and the Sindh Occupational Safety and Health Act/Rules — exact current provisions to be confirmed by the site legal/HSE function [S-06].	Heatstroke can be fatal or cause permanent organ damage; multiple casualties possible on peak days. Persons at risk: All outdoor workers, especially new arrivals, turnaround labour and PPE-intensive trades.	PARTIALLY VERIFIED: Sindh factory law requires ventilation/reasonable temperature, cooled drinking water for larger factories and rest provisions [S-06]; the site's own heat-stress programme (WBGT, work/rest, acclimatisation) is NOT evidenced.	4 x 4 = 16 — HIGH	ENGINEERING: shaded rest areas at each workfront; cooled drinking water and electrolytes at the point of work. ADMIN: site heat-stress procedure with measured-condition (e.g. WBGT) work/rest cycles, acclimatisation for new/returning workers, rescheduling of heavy work away from peak heat where practicable, buddy checks, supervisor training to recognise heat illness, rehearsed rapid-cooling and medical-evacuation response. PPE: breathable selections where the task allows.	HSE Manager / Contractor Supervisors Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 4 = 8 — MODERATE	Site heat-stress procedure; Sindh legal register [SITE INPUT REQUIRED] Review trigger / reference: Review at season start and after any heat-illness case. Refs: Sindh Factories Act 2015 [S-06]; [S-03].	Evidence status: Partially evidenced — site arrangements to be verified STOP-WORK: any suspected heatstroke case stops the crew's work, triggers medical response and a programme review; supervisors must stop work when the site procedure's measured limits are reached.
RA-22	Site traffic, mobile plant and material movement	Site roads, laydown and warehouse areas [CONFIRM traffic plan]	Vehicle–pedestrian collision, reversing in blind spots, load shift during internal transport of equipment/catalyst/bagged urea.	Fatality or serious injury to pedestrians, drivers and passengers. Persons at risk: Pedestrians in plant areas, drivers, passengers and other road users.	TO BE VERIFIED: traffic management plan, driver authorisation and journey management not reviewed.	4 x 5 = 20 — EXTREME	ENGINEERING: segregated walkways/crossings, physical separation in laydown areas, reversing alarms/cameras where fitted. ADMIN: site traffic plan (speed limits from the plan — not assumed), authorised-driver register, banksman for reversing in congested areas, load securing checks, seat-belt rule for all occupants. PPE: high-visibility clothing in traffic areas.	Construction Manager / Logistics Coordinator Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Traffic management plan; driver authorisation register Review trigger / reference: Review monthly and after any traffic incident. Ref: site traffic plan [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: any vehicle defect affecting brakes/steering/alarms removes it from service; pedestrian entry into an active movement zone stops the movement.
RA-23	Chemical and product handling beyond ammonia: SDS-dependent products, urea dust and bagging	Stores, bagging and chemical-use areas [CONFIRM product list]	Exposure to, fire from, or spill of actual site products (coatings, solvents, treatment chemicals, lubricants) whose identities are NOT public; urea dust and manual handling in bagging/despatch.	Occupational illness, fire, spill contamination, musculoskeletal injury. Persons at risk: Storekeepers, applicators, bagging/despatch crews and nearby workers.	NOT CONFIRMED: no product list or SDS file was available to this pre-assessment.	3 x 3 = 9 — MODERATE	VERIFY FIRST: full product list + current SDS for every product before mobilisation. ENGINEERING: ventilated mixing/application areas, approved flammables storage, bunded chemical storage, mechanical handling aids for bagged product. ADMIN: product-specific (COSHH-style) assessment per SDS briefed in the crew's language, spill kits at use points, waste via the site waste system. PPE: strictly per	HSE Officer / Contractor Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	SDS file; product-specific assessments Review trigger / reference: Review on every new product. Ref: product SDS [SITE INPUT REQUIRED].	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no product may be used before its SDS assessment is complete and briefed.

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
							SDS.				
RA-24	Emergency response interface: ammonia release, fire or plant emergency during occupancy	Whole defined site boundary	Workforce (incl. turnaround/contractor personnel) not aligned with the plant emergency plan — wrong muster point, missed headcount, downwind evacuation through an ammonia cloud, or blocked emergency access during an alarm.	Multiple fatalities in a major ammonia/fire event; delayed rescue. Persons at risk: Every person inside the boundary, especially new/transient workers; off-site community interface to be confirmed from the site plan.	TO BE VERIFIED: the approved site emergency plan, alarm signals, muster points, wind-direction rule and headcount system were NOT available — all are critical site inputs, none assumed here.	2 x 5 = 10 — HIGH	ADMIN: every worker inducted on actual alarm signals, muster points and the wind-direction/upwind rule from the approved plan before starting work; contractor headcount reconciled with the plant muster system (badge/gate data — Section 5); joint drills including contractors at the frequency set by the plan; ammonia-release drills with upwind evacuation; emergency access audited weekly; wind socks maintained at workfronts. ENGINEERING: alarm coverage audible/visible at all workfronts incl. high-noise areas.	HSE Manager / Plant Emergency Coordinator [title to confirm] Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved site emergency plan; drill and induction records Review trigger / reference: Review after every drill/alarm/workforce change. Ref: site emergency plan [SITE INPUT REQUIRED — critical].	Evidence status: To be verified — no site evidence reviewed HOLD POINT: no workfront may operate outside alarm coverage or without a briefed muster route; a failed drill triggers corrective action before the next high-risk activity.
RA-25	Occupational health exposures: dust, noise, vibration, welding fume and manual handling	Civil, mechanical, fabrication and urea areas [CONFIRM hygiene survey]	Respirable dust (incl. silica from concrete work), excessive noise, hand-arm vibration and repetitive manual handling causing chronic ill health; no exposure measurements exist in public sources.	Silicosis/lung disease, noise-induced hearing loss, vibration injury, musculoskeletal disorders (chronic). Persons at risk: Operators and nearby workers exposed repeatedly over shifts.	TO BE VERIFIED: no monitoring data in public sources — industrial-hygiene survey required before ratings can be confirmed.	4 x 3 = 12 — HIGH	ENGINEERING: wet cutting/on-tool extraction, low-noise/low-vibration tool selection, noise barriers/distance. ADMIN: competent hygiene survey and monitoring programme, signed hearing-protection zones from results, tool-time limits/rotation, health surveillance for exposed groups, task manual-handling assessments with mechanical aids. PPE: fit-tested RPE and hearing protection selected from monitoring results.	HSE Manager / Occupational Hygienist Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	Industrial hygiene survey and monitoring plan Review trigger / reference: Review on monitoring results. Ref: site hygiene plan [SITE INPUT REQUIRED].	Evidence status: To be verified — no site evidence reviewed STOP-WORK: dry cutting of concrete/stone prohibited where wet/extraction controls are unavailable.
RA-26	Nitrogen / inert-gas purging and blanketing — VERIFICATION-CRITICAL (only if used)	Systems under purge/blanketing [CONFIRM whether N2 is used]	Nitrogen leakage/discharge into work areas or vessels creating an oxygen-deficient atmosphere with no warning properties (colourless, odourless). Inerting is a standard ammonia-plant practice as a process type; Daharki-specific use is NOT public.	Rapid asphyxiation; fatalities including rescuers. Persons at risk: Personnel near vents, open ends, low areas and vessel entrants.	NOT CONFIRMED: nitrogen use at the workfront is not evidenced in public sources — confirm from commissioning/maintenance procedures.	2 x 5 = 10 — HIGH	VERIFY FIRST: confirm purging medium and procedure. ENGINEERING: discharge routed to a safe location; barricaded vent points. ADMIN: purge plan with marked exclusion zones, oxygen monitoring at access points, warning signage in workforce languages, prohibition on entering purged systems until oxygen is proven normal by test. PPE: personal O2 monitor where the procedure requires.	Commissioning Lead / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Purge/commissioning procedure; gas test records Review trigger / reference: Review at each purge operation. Ref: procedure [SITE INPUT REQUIRED].	Evidence status: NOT CONFIRMED — verification-critical HOLD POINT: no entry into, or breaking into, any purged system until oxygen content is proven acceptable by test and recorded.

NOTE: ammonia and other major-accident rows (RA-01/02/04/10) retain HIGH residual scores because severity cannot be reduced — only likelihood can. They are acceptable, if at all, only through the owner's formal acceptance route with all stated controls verified.

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8. Critical controls / verification checklist

A barrier is credited only when its evidence exists and a competent verifier has signed it. 'To be verified' = not yet credited.

#	Critical control / barrier	Applicability	Evidence required	Verifier (role)	Frequency	Verification status
1	Ammonia detection coverage verified for the work area; personal monitors issued where the site standard requires	To be verified	Detector layout/calibration records; monitor issue log	HSE Manager	Before ammonia work; calibration per schedule	Not verified
2	Process isolation / positive isolation proven and LOTO applied before intrusive work	To be verified	Isolation certificate with zero-energy verification	Authorised Isolator	Every intrusive task	Not verified
3	Line-opening controls: drain/vent/purge proven, first-break technique, gas test at break point	To be verified	Line-opening permit + gas test certificate	Mechanical Supervisor	Every line break	Not verified
4	Gas composition data received; CO/toxic-gas testing decision recorded (RA-08)	To be verified — composition data required	Composition data sheet; signed decision note	Process Safety / HSE Manager	Before intrusive work	Not verified
5	Catalyst SDS file complete and change-out assessment briefed (RA-07)	To be verified — SDS required	SDS file; assessment and briefing records	Process Engineer	Before catalyst handling	Not verified
6	Hot-work permit, ignition control, trained fire watch and post-work fire check	To be verified	Hot-work permit; fire-watch log; extinguisher checks	Hot Work Supervisor	Every hot-work task	Not verified
7	Confined-space controls: isolation, testing, attendant, communications, rescue equipment at entry point	To be verified	Entry permit; test records; rescue drill records	Entry Supervisor	Every entry	Not verified
8	Lifting controls: approved lift plan, certified crane/rigging, ground verification, barricaded drop zone	To be verified	Lift plan; certificates; set-up sign-off	Appointed Person (Lifting)	Every complex/critical lift	Not verified
9	Work-at-height controls: collective protection, tagged scaffolds, verified anchors, rescue plan at workfront	To be verified	Scaffold register/tags; harness records; rescue plan	Scaffold Supervisor / HSE Officer	Daily checks; weekly inspection	Not verified
10	Excavation controls: permit, services clearance/scan, protective system, daily competent-person inspection	To be verified	Excavation permit; scan records; inspection log	Competent Person (Excavations)	Daily and after rain	Not verified
11	Electrical controls: authorised persons only, prove-dead recorded, temporary-power inspection	To be verified	Electrical permits; prove-dead records; inspection register	Authorised Electrical Person	Every task; monthly installation check	Not verified
12	Pressure-test controls: approved test pack, calibrated gauges, barricaded exclusion, single command	To be verified	Test pack; calibration certificates; boundary-clear record	Test Supervisor	Every test	Not verified
13	SIMOPS / impairment control: permit coordination, no unapproved bypass of detection/ESD	To be verified	SIMOPS review record; impairment register	Area Authority	Every shift while active	Not verified
14	Emergency readiness: ammonia-release alarm, upwind muster, headcount reconciliation, joint drills	To be verified — emergency plan is a critical site input	Induction/drill records; reconciliation reports	HSE Manager	Drills per plan; headcount every shift	Not verified
15	Heat-stress controls: water/shade/rest at workfronts, acclimatisation, measured work/rest, heatstroke response	Site programme to be verified — Sindh legal duties noted [S-06]	Heat-stress procedure; monitoring logs; training records	HSE Manager	Daily in hot season	Not verified
16	Contractor competency: verified authorisations, language-appropriate induction/toolbox talks, supervision	To be verified	Competence matrix; induction records	Construction Manager / HSE Manager	At mobilisation; monthly audit	Not verified

9. Task sequencing / JSA / permit matrix

Permit names are generic functional descriptions. The owner's current PTW system governs actual permit types, forms and approval chain — obtain the controlled procedure (Section 14) and re-map this table before use. A task JSA/JHA must be produced per activity step and briefed before work.

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Task / activity	Prerequisite permit(s)	Isolation requirements	Drawings / documents	Competent persons	Pre-job briefing	Field verification	Return to service / handover
Ammonia system work / line opening	Ammonia work + line-opening permits; PTW	Positive isolation; drain/vent/purge proven	P&ID; line list; isolation certificate; gas test	Authorised Isolator; Gas Tester	First-break briefing incl. upwind escape route	Gas test + two-person line ID before break	Reinstatement checklist; leak test; operations acceptance
Hot work near process areas	Hot-work permit; PTW	Equipment isolation as required	Gas test certificate; fire-watch assignment	Hot Work Supervisor; Fire Watch	Permit briefing at work point	Gas test before/during; fire-watch log	Post-work fire check signed; permit closed
Confined-space entry	Entry permit; PTW	Blinding/isolation of all connected lines	Entry plan; rescue plan; test records	Entry Supervisor; Attendant; Rescue Team	Entrant/attendant briefing; comms check	Continuous monitoring; entry log	All persons accounted for; permit closed
Catalyst change-out	Entry + catalyst work permits	Vessel isolation/blinding; inerting per procedure	Catalyst SDS; change-out procedure	Process Engineer; Entry Supervisor	SDS-specific briefing	Atmosphere + dust controls verified	Spent-catalyst disposal record; vessel handback
Complex / critical lifting	Lifting permit (if site system requires)	Area cleared; plant isolation if over live plant (operations decision)	Lift plan; crane/rigging certs; ground assessment	Appointed Person; Operator; Riggers	Lift briefing incl. abort signals	Trial lift; drop-zone and weather check	Completion report; defects logged
Pressure testing	Pressure-test permit	Test-boundary isolation; blinds per test pack	Approved test pack; calibration certs	Test Supervisor; Mechanical Engineer	Boundary briefing; exclusion rules	Boundary-clear confirmation before pressurisation	Depressurisation record; test certificate
Excavation near services	Excavation permit	Service isolation where owner requires	Services drawings; scan records; excavation design	Competent Person (Excavations)	Briefing on marked services	Daily inspection; hand-dig zone enforced	Backfill/reinstatement record
Commissioning / energisation	Commissioning permit; energisation notice	Boundary isolation per certificate	PSSR records; boundary certificates; punch list	Commissioning Lead; Operations Rep	Boundary meeting, all affected parties	Permit cross-check before energisation	Operations acceptance certificate

10. Responsible-role matrix

Accountability is assigned to ROLES, never to invented individuals. Titles are functional descriptions; the site organisation chart and PTW procedure must confirm actual titles and authority.

Role	Accountability in this assessment	Authority / confirmation note
Area Authority / Operations Representative	Owns the plant area; authorises work and SIMOPS; first contact for plant upsets	Confirm against site org chart and PTW procedure
Permit Issuer / Receiver (per site PTW system)	Issue, acceptance, suspension and close-out of permits within competence	Exact titles per controlled PTW procedure — not assumed
Construction / Turnaround Manager	Delivery, resources, sequencing, interface management	Approves high-risk scheduling with HSE and operations
Contractor Supervisor (each contractor)	Direct crew supervision; toolbox talks; first-line control verification	One named supervisor per crew per shift — from mobilisation records
HSE Manager / HSE Officer	RA ownership, inspection, heat-stress and hygiene programmes, drill coordination	Stop-work authority to confirm in the site HSE plan
Process Safety Engineer	Major-accident interface, MOC/impairment, PSSR participation	Confirms where HAZID/HAZOP/LOPA studies are required
Mechanical Engineer / Supervisor	Piping, line opening, pressure testing, joint and equipment integrity	Approves test packs jointly with commissioning
Electrical / Instrumentation Engineer	Electrical safety, isolations, loop testing, energisation readiness	Maintains authorised-person register
Lifting Supervisor / Appointed Person	Lift planning, categorisation, set-up verification, execution control	Appointed in writing per owner lifting standard
Authorised Gas Tester	Atmospheric testing for permits, entries and hot work	Authorisation and calibration records current
Confined-Space Attendant / Rescue Team	Entry control, communications, rescue execution	Trained team + equipment at the entry point
QA/QC Inspector	Hold-point inspection: joints, welds, reinstatement	Releases ITP hold points only
Commissioning Lead	Completion, PSSR, energisation and handover boundaries	Single accountable authority at the construction/operations boundary
Emergency Response Coordinator	Alarm response, muster/headcount reconciliation, drills with plant operations	Interface defined in the approved emergency plan

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Role	Accountability in this assessment	Authority / confirmation note
[site title to confirm]		

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11. Action tracker

No action closes on wording alone — closure requires the stated evidence and closure-reviewer sign-off. Overdue Critical actions block the linked work.

Action ID	Linked RA	Action (specific)	Priority	Accountable role	Target	Verification evidence	Status	Closure reviewer
ACT-01	RA-01/04	Obtain P&IDs, ammonia inventory and operating data; fix boundary, workfronts and asset IDs; re-issue RA against them	Critical	Operations / Document Control	Before approval	Plot plan + data on file; RA Rev. 1	Open	HSE Manager
ACT-02	RA-07/08	Obtain stream-composition data and full SDS file (incl. catalysts); record RA-07/08 control decisions	Critical	Process Safety Engineer	Before intrusive work	Data/SDS on file; signed decision notes	Open	HSE Manager
ACT-03	All	Replace PROPOSED matrix with owner matrix; re-rate all 26 entries in a validation workshop	Critical	HSE Manager (owner)	Before approval	Workshop record; re-rated register	Open	Project Manager
ACT-04	All	Obtain authorized owner HSE/PSM/PTW/LOTO/hot-work/confined-space/line-opening/lifting/pressure-test/ MOC/emergency procedures; re-map Sections 8-9	Critical	Document Control (owner)	Before approval	Controlled copies in register	Open	HSE Manager
ACT-05	RA-24	Obtain approved emergency plan; extract alarm, upwind muster, medical and rescue provisions into induction	Critical	HSE Manager	Before work starts	Plan on file; induction updated	Open	Operations Representative
ACT-06	RA-24 / Sec.5	Start daily actual-headcount capture (gate/badge + contractor returns); reconcile with muster; populate Section 5.2	Critical	Contractor Admin / Security	Before work starts; then daily	Dated manpower reports; reconciliation log	Open	HSE Manager
ACT-07	RA-09/10	Verify isolation hardware, blind-register practice and authorised-isolator appointments by field inspection	Critical	Area Authority	Before first line break	Inspection report; written appointments	Open	Mechanical Engineer
ACT-08	RA-13	Complete ground-bearing assessments for planned critical-lift set-up positions	High	Appointed Person (Lifting)	Before first critical lift	Signed set-up verifications	Open	Construction Manager
ACT-09	RA-21	Confirm heat-stress programme: Sindh legal duties, monitoring, water/shade/rest, acclimatisation roster	Critical	HSE Manager	Before hot season / immediately	Programme document; monitoring logs	Open	Construction Manager
ACT-10	RA-15	Complete underground-services survey/scan for first-package excavation areas	Critical	Civil Supervisor / Operations	Before excavation	Scan records; marked drawings	Open	Construction Manager
ACT-11	RA-11	Prove confined-space rescue capability by practical drill at the first entry location	Critical	Entry Supervisor	Before first entry	Drill report with corrective actions	Open	HSE Manager
ACT-12	RA-17	Approve test packs (hydrostatic default; pneumatic only with engineered justification); verify gauge calibration	High	Mechanical / Commissioning Engineer	Before first test	Approved packs; certificates	Open	Commissioning Lead
ACT-13	RA-25	Commission industrial-hygiene baseline survey (dust/silica, noise, vibration, fume)	Moderate	HSE Manager	Within first month	Survey report; monitoring plan	Open	HSE Manager
ACT-14	RA-01	Verify ammonia detector coverage, calibration and alarm response for interface areas by field check	Critical	HSE Manager / Operations	Before ammonia work	Field check report; calibration records	Open	Process Safety Engineer
ACT-15	All	Conduct site walkdown and worker consultation; record observations and amend the register	Critical	HSE Manager	Before approval	Walkdown report; consultation records; RA Rev. 1	Open	Project Manager

12. Emergency response and rescue interface

The approved site emergency plan was NOT available. Nothing below is assumed: every operational detail is a critical site input to be extracted from that plan and briefed at induction (RA-24, ACT-05).

Emergency element	Detail required from the approved site emergency plan	Status
Emergency number / control room	Exact numbers and radio channels from any workfront	SITE INPUT REQUIRED — critical
Alarm signals	Fire, ammonia/gas release, evacuation and all-clear signals; audibility at high-noise workfronts	SITE INPUT REQUIRED — critical
Muster points	Locations per area with UPWIND alternates and routes from every workfront	SITE INPUT REQUIRED — critical

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Emergency element	Detail required from the approved site emergency plan	Status
Headcount reconciliation	Badge/gate system, who reconciles contractor numbers, reporting line to incident command	SITE INPUT REQUIRED — critical
Wind indication	Wind-sock locations and the upwind-muster selection rule for ammonia releases	SITE INPUT REQUIRED — critical
Medical provision	Clinic location [CONFIRM], ambulance provision and hospital route — no response times stated until evidenced	SITE INPUT REQUIRED — critical
Fire response	Plant fire-team interface, firewater status at interface, construction-area extinguisher provision	SITE INPUT REQUIRED — critical
Ammonia-release response	Detector-alarm actions, safe refuge, respiratory escape provision, re-entry authority	SITE INPUT REQUIRED — critical
Rescue — confined space / height	Teams, equipment and drill standards (RA-11/RA-14)	SITE INPUT REQUIRED — critical
Contractor-operator interface	Who commands, who accounts for contractor personnel, permit suspension on alarm	SITE INPUT REQUIRED — critical

13. Environmental and occupational-health screening

Screening only: no exposure level is stated anywhere, because no monitoring data exists in public sources. Credible exposures raise a hygiene action (ACT-13).

Aspect	Source activity (register link)	Screening conclusion	Control / monitoring requirement
Ammonia emissions / release	Ammonia plant, storage, transfer (RA-01/02)	Major-accident + routine-emission topic — unquantified	Detector coverage; leak detection/repair programme; emergency plan interface
Dust / silica; urea dust	Concrete work; urea handling (RA-06/15/25)	Credible exposure — unquantified	Wet cutting/extraction; hygiene survey; fit-tested RPE where required
Welding fume; noise; vibration	Welding, tools, turbines (RA-12/20/25)	Credible exposure — unquantified	LEV; noise/vibration survey; health surveillance
Heat stress	All outdoor work (RA-21)	HIGH — Sindh jurisdiction [S-06]	Site heat programme; water/shade/rest; acclimatisation; medical response
Spills / effluent / waste	Refuelling, chemicals, turnaround waste (RA-23)	Credible — consequence-driven	Bunding, spill kits, drain protection, licensed waste route per current Pakistan/Sindh requirements [CONFIRM]
Water use and sanitation	Welfare provision (Sec. 5/14)	Welfare-critical in heat	Potable water at workfronts; sanitation/rest facilities verified at mobilisation [CONFIRM]

14. Document limitations, assumptions and site-data request list

LIMITATIONS — BLUNT STATEMENT. This document is a publicly researched pre-assessment based solely on the Section 17 sources (accessed 28 October 2026) and established HSE practice for ammonia/urea plants. It has NOT been verified by a walkdown, operator/worker interview, approved drawings, current permits, process safety information or a shift headcount. It is not an official Engro document, not Engro-approved, not a method statement, and not a substitute for the controlled owner HSE/PSM system, task JSAs or specialist studies (HAZID/HAZOP/LOPA/QRA). Public information on Daharki is thin: no plot plan, inventory, P&ID, detector layout, contractor register or current headcount is public. Any person treating this draft as authorization to work would be misusing it.

ASSUMPTIONS (all flagged for confirmation): (1) boundary = Daharki complex (Base Plant + EnVen + shared utilities/storage) only; Zarkhez Port Qasim excluded; (2) the activity list reflects the publicly evidenced plant types and the 2024 turnaround scope types plus standard live-plant interface tasks — each row applies only where the approved work pack confirms it; (3) the PROPOSED 5x5 matrix is used because the owner matrix was unavailable; (4) permit/role names are generic functional descriptions.

14.1 Site-data request list

#	Site input required	Why it matters	Owner role	Priority
1	Approved plot plan, exact boundary, unit/workfront and asset IDs (Base Plant, EnVen, utilities, storage)	Defines what this RA covers; every area reference stays [CONFIRM] without it	Operations / Document Control	Critical — before approval
2	Current work package: task list, method statements, JSA/JHA set and look-ahead schedule	Confirms which register rows apply and which are N/A	Construction Manager	Critical — before approval
3	P&IDs, line/equipment lists; ammonia inventory; design/operating pressures, temperatures	Required for RA-01/02/04/09/10/17; no values are assumed in this draft	Process / Mechanical Engineer	Critical — before intrusive work

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#	Site input required	Why it matters	Owner role	Priority
4	Gas/stream composition data and full SDS file (process streams, catalysts, products)	Decides RA-07/RA-08/RA-23 controls; verification-critical hazards cannot close without it	Process Safety Engineer	Critical — before intrusive work
5	Owner risk matrix and risk-acceptance criteria	Replaces the PROPOSED 5x5 matrix in Section 6	HSE Manager (owner)	Critical — before approval
6	Current owner HSE/PSM procedures: PTW, LOTO, hot work, confined space, line opening, excavation, lifting, pressure testing, MOC, ammonia emergency response	Governing documents for execution; not publicly available and not reviewed for this draft	Document Control (owner)	Critical — before approval
7	Approved emergency plan: alarm signals, muster points, upwind rule, medical/ambulance, ammonia-release and off-site interface	RA-24 cannot close without it; no locations/response times assumed	HSE Manager / Emergency Coordinator	Critical — before work starts
8	Actual headcount by shift: gate report, contractor mobilisation register, daily manpower report	Current total inside the boundary is not publicly disclosed; muster depends on it	Contractor Admin / Security	Critical — before work starts; then daily
9	Hazardous-area drawings, electrical single-lines, underground-service drawings/surveys	Govern hot work, temporary power, excavation and crane set-up	E&I / Civil Engineers	Critical — before hot/electrical/excavation work
10	Tank, boiler, reformer and rotating-equipment inspection/integrity records	Close RA-02/03/19/20 hold points for actual equipment condition	Mechanical / Inspection Engineer	High — before turnaround/intrusive work
11	Current-season heat-stress arrangements and Sindh legal register confirmation	Confirms RA-21 controls and exact Sindh duties for the work dates	HSE Manager	High — before hot season
12	Site walkdown, worker consultation, incident/near-miss history and inspection findings authorised for this RA	Grounds the RA in observed conditions, not public sources alone	HSE Manager	High — before approval

15. Approval, validation and review

15.1 Implementation, validation & review

Item	Requirement for this assessment
Critical controls / hold points	Section 8 controls and each Section 7 hold point must be verified with evidence before the associated work starts. No intrusive ammonia work without the RA-01/RA-08 data decisions recorded.
Pre-start verification	PTW active; isolations proved; composition/SDS decisions recorded (ACT-02); owner matrix adopted (ACT-03); emergency arrangements briefed; toolbox talk done; headcount capture running (ACT-06).
Review triggers	Section 15.3 — any trigger requires re-validation of affected rows before work continues.
Emergency arrangements	All operational details are critical site inputs from the approved plan (Section 12). Nothing here states a muster point, response time or alarm level.

15.2 Consultation & approval

Prepared by / role / date	Operations / supervisor review	HSE review	Authorized approver / date
Role: HSE — competent assessor Name: _____ Signature: _____ Date: _____	Role: Area Authority / Operations Representative (Construction Manager review also required) Name: _____ Signature: _____ Date: _____	Role: HSE Manager Name: _____ Signature: _____ Date: _____	Role: as defined by the owner's document system Name: _____ Signature: _____ Date: _____

Approval block intentionally left blank: names, signatures and dates may only be entered by real authorized persons after site validation. Signing this Rev. 0 without closing the Critical inputs in Section 14.1 would not constitute a valid approval for execution.

15.3 Review triggers

Change in scope, method, equipment or process status; new SIMOPS condition; any incident or significant near miss; change in composition or isolation boundary; new contractor or major workforce change; onset of peak heat season or a severe-weather event; failed critical control; expired permit conditions; or the review interval in the owner's procedure — whichever occurs first.

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16. Quality check

QA check	Result
All numeric figures carry a source/date/scope label	PASS — Sections 3 and 5 tie every figure to [S-01..S-05] with scope notes
No corporate/turnaround figure presented as current Daharki headcount	PASS — Section 5.1 warning; Section 5.2 left as a site-record form
No invented site count, workfront, process condition, inventory, incident statistic, permit/drawing number, name, signature or approval	PASS — unknowns are [CONFIRM]/[SITE INPUT REQUIRED]; approval block blank
Daharki vs Port Qasim (Zarkhez) not confused	PASS — Zarkhez recorded as a separate Karachi site and excluded (Sections 3, 4)
Base Plant capacity discrepancy handled honestly	PASS — PSX-notice figure 950,000 t used; 975,000 t secondary figure flagged in [S-04]
Every High/Extreme hazard has controls, a responsible role, verification evidence and a hold point	PASS — checked row by row in Section 7; mirrored in the XLSX register
Risk scores consistent with the stated matrix; no residual reduction without credible controls	PASS — scores computed from the PROPOSED matrix; severity not reduced on major-accident rows
'Existing control' vs 'additional action' distinguished	PASS — separate columns; existing controls carry evidence status
Legal requirements vs technical guidance not misclassified; Pakistan jurisdiction used	PASS — Sindh law labelled LAW; ISO labelled STANDARD; NIOSH/OSHA labelled US GUIDANCE only; no Saudi rules imported
Source links and claim-level citations present; thinness flagged	PASS — Section 17; thinness note in Section 3 and Section 14
Tables legible; landscape registers; repeated headers; footer with doc ID, Rev. 0, NOT APPROVED FOR EXECUTION and page numbers	PASS
Status states 'not approved for execution' on cover and footers	PASS

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17. References and traceability

Access date for all sources: 28 October 2026. Types: [LAW] Pakistan/Sindh legal, [CLIENT] owner-specific, [STANDARD] international standard, [GUIDANCE] technical guidance only.

Ref	Source (title — organisation, date)	Exact URL	Assertion(s) supported in this document	Verification status
S-01	World's largest Urea Plant starts production in Pakistan (press release, 4 Jan 2011) — Engro Fertilizers Limited, 2011-01-04	https://www.engrofertilizers.com/trust/press-release-detail/2011/worlds-largest-urea-plant-starts-production-in-pakistan/	EnVen at Daharki is a single-train ammonia-urea plant, capacity 1.3 million tons/yr; took Engro Fertilizers' total urea capacity to 2.3 million tons/yr; USD 1.1bn investment; described as Pakistan's most energy-efficient fertilizer plant at launch. Official operator source.	Verified — opened 28 Oct 2026
S-02	Engro Fertilizers resumes operations of EnVen plant (PSX notice reported 10 Jun 2025) — Business Recorder, 2025-06-10	https://www.brecorder.com/news/40366968/engro-fertilizers-resumes-operations-of-enven-plant	EnVen (Daharki, Sindh; established 2011; 1.3 Mt/yr) resumed operations 7 Jun 2025 after unscheduled maintenance (shut 5 Jun 2025). Engro Fertilizers Ltd incorporated 29 Jun 2009, wholly owned subsidiary of Engro Corporation; manufactures/markets fertilizers, seeds, pesticides. Proves the complex was operating in 2025 — current Oct 2026 status to be confirmed with the operator.	Verified — opened 28 Oct 2026
S-03	EnVen Plant maintenance completed with investment of $50m (22 Jun 2024) — Business Recorder, 2024-06-22	https://www.brecorder.com/news/40309272/enven-plant-maintenance-completed-with-investment-of-50m	55-day scheduled EnVen turnaround completed 18 Jun 2024; ~USD 50m; >5,000 technical activities incl. first waste-heat-boiler replacement, furnace convection-section overhaul, ammonia storage tank inspection, overhaul of six major turbo trains and gas turbines; >6,000 people at peak ON THAT TURNAROUND (historical, turnaround-specific — NOT current headcount); completed 'despite extreme weather conditions'.	Verified — opened 28 Oct 2026
S-04	Engro Fertilizers increases base plant urea capacity to 950,000 tons (PSX notice reported 29 Dec 2022) — Pakistan Today (Profit) / PSX notice, 2022-12-29	https://profit.pakistantoday.com.pk/2022/12/29/engro-fertilizers-increases-base-plant-urea-capacity-to-950000-tons	Base Plant (Plant-1) at Daharki capable of up to 950,000 tons urea/yr after maintenance/debottlenecking (previously 780,000, before that 500,000); two urea plants at Daharki (Base + EnVen) and Zarkhez NPK at Port Qasim, Karachi (separate site, excluded). Note: one secondary summary states total Daharki urea capacity 2.275 Mt with Base at 975,000 t — this assessment uses the PSX-notice figure of 950,000 t and flags the discrepancy for operator confirmation.	Verified — opened 28 Oct 2026
S-05	Engro Fertilisers Daharki Ammonia Plant (plant entry) — Global Energy Monitor (GEM.wiki), Entry current at access date	https://www.gem.wiki/Engro_Fertilisers_Daharki_Ammonia_Plant	Daharki ammonia plant, Sindh: owner Engro Fertilisers Limited (100%); primary product ammonia, secondary products incl. urea; feedstock natural gas; status operating. Independent corroboration of operator, products and feedstock only — no site conditions taken from it.	Verified — opened 28 Oct 2026
S-06	Sindh Factories Act 2015 (Act XIII of 2016) and Sindh OSH framework — heat/ventilation, drinking water, hours — Government of Sindh, 2015/2016 (as amended)	http://sindhlaws.gov.pk/setup/publications_SindhCode/PUB-NEW-26-000073.pdf	[PAKISTAN / SINDH LAW] Factory duties incl. adequate ventilation and reasonable temperature, cooled drinking water in hot months for larger factories, rest provisions, and working-hours limits. Cited at topic level only — exact current provisions/applicability to be confirmed by the site legal/HSE function. Pakistan has no Saudi-style statutory midday-work ban; do not import one.	Verified — source text checked 28 Oct 2026
S-07	Ammonia — NIOSH Pocket Guide / IDLH documentation (technical guidance) — US NIOSH / CDC, Current at access date	https://www.cdc.gov/niosh/npg/npgd0028.html	[GUIDANCE — US, not Pakistan law] Ammonia REL 25 ppm TWA, STEL 35 ppm; IDLH 300 ppm. Used only to characterise ammonia toxicity as a process-type hazard; site alarm/action limits must come from the approved site standard.	Cited from NIOSH data via search/open check 28 Oct 2026 — verify on NIOSH page before site use
S-08	Ammonia — OSHA permissible exposure limit (technical guidance) — US OSHA, Current at access date	https://www.osha.gov/chemicaldata/62	[GUIDANCE — US, not Pakistan law] Ammonia PEL 50 ppm 8-hr TWA. Technical reference only.	Cited 28 Oct 2026 — guidance only
S-09	Process Safety Management — Overview / Hazard Recognition (technical guidance) — US OSHA, Current at access date	https://www.osha.gov/process-safety-management	[GUIDANCE — US] PSM concepts structuring major-accident interface controls (RA-04/08/18). Not Pakistan law; owner PSM system governs.	As cited 28 Oct 2026
S-10	Confined Spaces — Overview (technical guidance) — US OSHA, Current at access date	https://www.osha.gov/confined-spaces	[GUIDANCE — US] Confined-space control concepts supporting RA-11.	As cited 28 Oct 2026
S-11	Welding, Cutting and Brazing — Hazards and Solutions (technical guidance) — US OSHA, Current at access date	https://www.osha.gov/welding-cutting-brazing/hazards-solutions/	[GUIDANCE — US] Hot-work/welding-fume control concepts supporting RA-12.	As cited 28 Oct 2026
S-12	ISO 45001:2018 + Amd 1:2024 — OH&S management systems (framework) — ISO, 2018, confirmed 2024	https://www.iso.org/standard/63787.html	[STANDARD] Management-system framework for this assessment's process. Not a hazard checklist and not Pakistan law.	Verified status 28 Oct 2026

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Requirement-type key: Sindh factory/OSH law [S-06] is Pakistan LAW. Engro owner procedures (not reviewed) would be CLIENT requirements. ISO 45001 [S-12] is a STANDARD used as framework. NIOSH/OSHA material [S-07..S-11] is US GUIDANCE, never presented as Pakistan law. END OF DOCUMENT — Rev. 0 — Public-source pre-assessment.

Template use note: Risk ratings, legal/client references, workforce data, inspection observations and approvals must be verified by competent and authorized personnel. The generic reference format used for layout does not establish compliance or owner acceptance.

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